

SPECIFICATION SHEET
Mountain Home AFB, ID
Acquisition of Particle Analysis System
Monday, February 23, 2026

1.0 GENERAL SCOPE

This document defines the technical and performance requirements for one (1) Malvern Panalytical Zetasizer Lab (Red) system. The vendor shall provide the system, including all necessary hardware, software, accessories, self-installation support, training, and warranty as detailed herein. The purpose of this acquisition is to establish the capability to analyze the size and stability (zeta potential) of microscopic particles and molecules within liquid dispersions.

2.0 BACKGROUND

Our engineering and quality control processes require precise characterization of colloidal systems. The stability and performance of these systems are directly related to particle size and surface charge. This acquisition will provide the essential capability to measure these parameters accurately, ensuring product quality and supporting research and development initiatives, particularly for materials in conductive media.

3.0 APPLICABLE DOCUMENTS & STANDARDS

The supplied system and all related services shall comply with the latest versions of the following standards, where applicable:

ISO 13321: Particle size analysis — Photon correlation spectroscopy (or latest equivalent ISO standard for DLS).

ISO 13099: Colloidal systems — Methods for zeta potential determination (or latest equivalent).

CSA (Canadian Standards Association): The system must be CSA validated, as specified in the quotation.

4.0 MINIMUM TECHNICAL REQUIREMENTS (SALIENT CHARACTERISTICS)

4.1 Performance Requirements

4.1.1. Primary Function: The system must be capable of measuring colloidal particle size, molecular weight, and zeta potential of dispersed particles and molecules.

4.1.2. Measurement Range: The system must meet the manufacturer's standard measurement ranges for the Zetasizer Lab (Red) model for size, zeta potential, and molecular weight.

4.1.3. Accuracy & Precision: System accuracy and precision shall be verifiable using the included Zeta Potential Transfer Standards.

4.1.4. Sample Conditions: The system must be optimized to perform high-quality measurements in conductive media.

4.1.5. Throughput: The system must incorporate a feature for "Adaptive Correlation" to provide faster, more reproducible size measurements.

4.2 Hardware Requirements

4.2.1. Light Source: Must be a 10mW HeNe laser operating at a 633nm wavelength.

- 4.2.2. Detector: Must be a high-sensitivity Avalanche Photodiode (APD) detector.
- 4.2.3. Proprietary Technology: Must include M3-PALS (Mixed-Mode Measurement - Phase Analysis Light Scattering) and a Constant Current Zeta Mode for zeta potential analysis, and Adaptive Correlation for size analysis.
- 4.2.4. Measurement Angle: Size and Molecular Weight measurements must be performed at a 90-degree scattering angle.

4.3 Software & Data Management Requirements

- 4.3.1. Operating System: Software must be compatible with standard, modern Microsoft Windows operating systems (Windows 10/11).
- 4.3.2. User Interface: Must include the ZS Xplorer Application Software with an intuitive graphical user interface (GUI), an easy-to-use method builder, and advanced data quality assessment tools.
- 4.3.3. Data Export: Must support data export in non-proprietary formats (e.g., .csv, .txt) for analysis in third-party software.
- 4.3.4. Updates: Free software updates must be provided for the lifetime of the instrument.

5.0 INSTALLATION AND ACCEPTANCE

- 5.1. Contractor Responsibilities: The contractor shall provide a complete "Self Install" package. This includes providing all required hardware, software (via download), and comprehensive, easy-to-follow installation instructions and digital training resources.
- 5.2. Government/Buyer Responsibilities: The buyer will provide a suitable computer and location for the self-installation.
- 5.3. Acceptance Testing: Acceptance will be considered complete upon successful self-installation of the hardware and software, and successful verification of system performance using the manufacturer-supplied Zeta Potential Transfer Standards according to the provided instructions.

6.0 TRAINING

The contractor shall provide a comprehensive, multi-format training program.

- 6.1. Operator Training (eLearning): The purchase must include access to the Zetasizer Advance Essentials eLearning Course.
- 6.2. Advanced Training: The purchase must include one (1) voucher for a single seat at one of the contractor's regularly scheduled classroom training courses.
- 6.3. Training Materials: All necessary digital training resources shall be provided for self-installation and operation.

7.0 DOCUMENTATION

The contractor shall deliver the following documentation in English at the time of delivery:

User & Operator Manuals (Digital and/or Hard Copy)

Software Installation Guide

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Self-Installation Instructions and Digital Resources

CSA Validation Documentation

8.0 WARRANTY & SUPPORT

8.1. Warranty Period: The system must include a comprehensive full one (1) year warranty covering all parts and labor, effective from the date of delivery.

8.2. Technical Support: The contractor must provide free helpdesk support via telephone and/or email for the lifetime of the analyzer to assist with troubleshooting and general system/application guidance.

9.0 DELIVERABLES CHECKLIST

1. The final delivery must include, at a minimum, the following line items:
 - 1.1. Zetasizer Advance Series - Lab (Red) System (P/N: ZSU3105): 1 Each
 - 1.2. Folded Capillary Zeta Cell (Box of 10) (P/N: DTS1070): 1 Box
 - 1.3. Zeta Potential Transfer STD Pack of 10 (P/N: ZTS1240): 1 Pack
 - 1.4. Zetasizer Advance Self Install Kit (P/N: ZSU8005): 1 Each
 - 1.5. Zetasizer Advance Essentials eLearning (P/N: ELC0020): 1 Each
 - 1.6. Training - 1 Seat (P/N: CTI2000): 1 Voucher

10.0 SHIPPING AND DELIVERY

10.1. Shipping Costs: All costs for shipping, freight, handling, and insurance must be included in the final quotation.

10.2. Delivery Address: 1300 Liberator Street, Mountain Home AFB, Idaho, 83648

10.3. Delivery Deadline: Delivery shall be made no later than 6 weeks after receipt of order (ARO).